Evaluation Of Cancer in World Countries

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Summary of the project:

Millions of lives are lost to cancer every year, making it one of the top causes of mortality globally. It is one of the biggest health issues facing the world today, impacting individuals everywhere. The main causes of this are hereditary vulnerability, exposure to the environment, and lifestyle decisions. Lung, skin, colon, prostate, and breast cancer are the most fatal varieties of the disease. Many people who are diagnosed with cancer may feel overwhelmed by their prognosis and the treatments that they must endure, frequently feeling scared and uneasy. The chance of developing cancer can also be raised by several risk factors. These include long-term stress, being subjected to physical or psychological abuse, bullying, or maltreatment, all of which increase a person's susceptibility to the illness overall. Cancer is a disease that can afflict anyone, despite certain demographics being more susceptible than others, such as men for prostate cancer or specific age groups for diverse types of cancer. The fight against cancer is continuing and requires constant attention and resources despite advances in medical research and treatment. Cancer is still a major worldwide health concern.

Cancer is an issue that is not treated with the urgency it requires, and it is not discussed enough, despite its significant relevance and impact on our society. As mentioned in this study, it is a disease that affects everyone globally for numerous reasons, including psychological, emotional, and economic impacts. Moreover, it is usually more common to hear about people who succumb to cancer, but the struggles of those living with the disease are not highlighted as much: there are also many people who battle cancer and survive but, due to the aggressive treatments and the toll the disease takes on their bodies, they are left with serious physical and mental challenges for the rest of their lives. These survivors often face ongoing health issues and are at higher risk for developing secondary cancers or other health complications.

For this reason, if cancer were to be thoroughly studied and properly analysed, it could potentially be prevented or, at the very least, treated more effectively. This can be achieved through various methods such as increasing access to early detection screenings, enhancing the availability of advanced treatment options, and promoting public awareness campaigns about lifestyle changes that reduce cancer risk. However, for these measures to be successfully implemented, it is crucial to begin by comprehensively studying the problem and recognizing the significant impact it has on our global society. Understanding the full scope of cancer's effects is essential for developing effective strategies to combat this pervasive disease.

Problem to be studied.

The general problem I have chosen to study is cancer disease in worldwide countries. I need to raise awareness on specific aspects of this disease that are not enough taken into consideration. In order to do this, my study is divided into the three parts described below.

First, the evolution of Cancer Patients around the world from 1990 to 2017 due to several reasons such as radiation or effects due to today's lifestyle. Afterwards, I focused on some countries in Europe such as Albania, Croatia, France, Germany, Spain, Andorra, Portugal, Poland, and Norway to get more detailed information about these changes. With this information I answered the following questions:

- How much has cancer diagnosis increased over the years?
- Which countries have the highest and the lowest cancer cases?
- Do the different countries in Europe have big differences in cancer cases?

In second place, I analysed the GDP of world countries, as this gives an idea of their wealth. The goal was then to compare the rates of cancer mortality in these countries with

GDP rates in order to draw conclusions about the role of wealth in cancer mortality. Some of the questions that I answered in this part were:

- Are cancer mortality rates linked to the wealth of countries?
- Does the increase of GDP necessarily lead to lower cancer mortality rates?
- How have cancer mortality rates evolved across time? Are these evolutions in accordance with GDP evolutions?

Finally, to understand the real impact of cancer, I aimed to discover if cancers of the colon, skin, breast, prostate, and lung are major causes of death worldwide, in comparison to other causes of death that receive more attention on social media and in our society in general, such as homicides, alcohol and drug abuse, and traffic fatalities. By examining these cancer types, I hoped to highlight their significant impact and the need for increased awareness and research to better address these prevalent health issues.

Objectives

The analysis of worldwide cancer has various goals. First, I wanted to identify the role that the wealth of countries has in cancer death rates. One of the objectives of my project is to analyse if countries' wealth is linked with cancer mortality rates. The hypothesis is that the higher GDP a country has, the lower cancer mortality rates it should have. Indeed, a successful economy with a high GDP means that the country has resources in order to research, prevent, and treat cancer diseases. The cancer mortality rates of countries will give me an overall overview of the efficiency of each country's combat against cancer. A low cancer mortality rate means that the country is successful in preventing and treating cancer cases. I have also researched how much cancer has varied throughout the years and how the different historical events such as Chernobyl have impacted the number of cancer patients and how some countries contribute to this cause. Finally, I aimed to analyse the impact of colon, skin, breast, prostate, and lung cancers in our society, in comparison to other causes of death such as traffic fatalities, homicides, and drug abuse. By doing so, I hoped to draw meaningful conclusions regarding the primary focus of my project. Through this analysis, I sought to understand the relative burden of these cancers and highlight the importance of prioritizing them in public health discussions and interventions.

Analysis Methodology

Data sources

The following are the data sources that we used:

- Link 1. I took the main data about cancer mortality from this source. "Our World in Data" provides information about the world's most pressing issues based on the findings and studies conducted by prestigious journalists, media outlets, academics, and research institutions**,** so I have no doubt on the veracity of the data. I used a publication providing numbers on cancer mortality in world countries from 1970 to 2017.
- Link 2, Link 3, Link 4, Link 5, Link 6. I used these other pages of data from "Our World in Data" to extract information regarding the cancer diagnoses and deaths worldwide, and to analyse and see the evolution of these rates over the years. All the information from these pages was then compared.
- Link 7.Link 8 The data about GDP levels were taken from "The World Bank." The World Bank is a globally recognised institution which publishes data in a wide range of topics. Their data is reliable and easily accessible to the public. I

downloaded the data giving us the GDP in US$ of world countries per year, between 1960 and 2022.

Analysed variables.

I started out by analysing the number of diagnoses each country has for each year, and how much the lowest and the highest country has increased over the 27 years that I studied. Afterwards I took the data of some European countries to get a closer look at the evolution of the countries near Spain.

Then, I analysed the GDP of world countries. The World Bank open data gave me the exact GDP of each world country between the years 1960 and 2021. I then converted this information into maps to have a more visual comprehension of it. This variable gives a clear idea of the world's wealthiest countries and how these have evolved over time.

Next, I analysed the cancer death rates and the unemployment rates of the same world countries. I attempted to figure out whether there was a relationship between these two variables.

Then, I analysed the number of fatalities due to alcohol per 100,000 individuals. The goal was to determine whether alcohol use and the quantity of cancer fatalities among the five most prevalent types of cancer were correlated in any way.

Lastly, I made a comparison between cancer and other causes of death. The number of homicides per hundred thousand individuals, the number of road deaths per 100,000 people, and the number of drug abuse-related fatalities per 100,000 people are the variables I used to calculate the ratio among cancer and the other causes of death.

Appropriate statistical analysis

The following was used to carry out the project's statistical analysis:

- Standard Deviation to study the change of cancer cases around the globe.
- Arithmetic means the number of cancer deaths of each type, and the other variety of deaths over time.
- Simple Index Number to compare the development of cancer deaths over time.
- The proportion of deaths that result from dividing the number of cancer deaths by the total number of deaths from other causes.
- Arithmetic means and quartiles of GDP to classify countries according to their wealth.

Type of visualization used.

I used choropleth maps to represent and summarize the GDP of world countries. The intensity of the colour represents the level of GDP: the higher intensity meaning the highest GDP. I used three of these maps to have a picture of GDP level of three different years and observe the evolution of these over time.

Furthermore, I used a pie chart (in this case an Excel file) to figure out the proportion of deaths between cancer deaths and homicides, among cancer deaths and traffic fatalities, as well as between cancer deaths and drug abuse. This is due to the pie chart being the most effective way to see the relationship between the two categories.

I also used pie charts to represent each percentage of the causes of cancer, and map graphs to show the diversity of cancer patients around the world. To represent the different changes in the cancer cases over time I used line graphs and bar charts to compare them to different countries around the world.

Pre-processing of the data

To be able to get a picture of how many cancer patients there are in each country I had to analyse the data that I obtained. To do this I filtered each year and represented it in a map chart to get more visual representation and I obtained the countries with the highest and the lowest number of cancer patients in 2017 since it was the most updated year. Afterwards I decided to focus on a few countries in Europe to get more detailed data since I had less numbers to work with and reached the main causes of this illness.

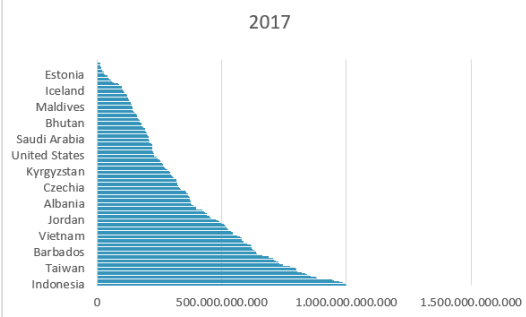
I first had to process the data to realize the choropleth map representing world countries' GDP. After downloading the data into an Excel file, I had to delete the irrelevant data. For example, between the list of countries' names, I found "European countries," "Lower income countries" or "South Asian countries" and their GDP. This information is irrelevant as I was only going to use the GDP of individual countries and not from a group of them. After deleting this information, I used the "filter" button and unchecked the "blanks" for the year 1960. I then created a map with the remaining data and repeated the process for the years 1990 and 2021.

Finally, to analyse the data to compare it to the other causes of deaths. To start with I took the number of deaths per cancer type, I did this by creating a dynamic table with bank of data of mortality rates. To be able to compare worldwide, I got the population from the data bank and did the same process as before. Next, I created three new variables from the data bank studying the deaths from homicide, alcohol and drug abuse and traffic fatalities. To end with, the other three variables were created by dividing one variable by another. The first of the three variables I created was Deaths from Cancer (colon, skin, breast, prostate, and lung) per Death due to Drug Abuse, then is Deaths from Cancer (colon, skin, breast, prostate, and lung) per Death due to Traffic Fatalities and the third variable is Deaths from Cancer (colon, skin, breast, prostate, and lung) per Homicide.

Statistical analysis

Cancer around the world:

I have focused on 2017 for this graph which is the most recent year that we are studying, were the average of cancer cases are 349.174.935.754 and the country with more cancer patients is Indonesia and the lowest Estonia, as we can see in the graph.

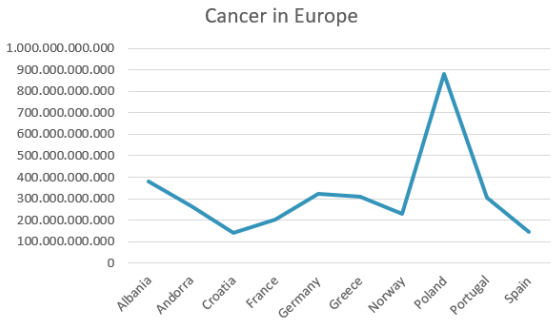


The next tool I used was to measure variability, in this case the Standard Deviation. The result I obtained was 2,48. This means the number of suicides between the different countries does not vary considerably.

Lastly, I am going to focus on a few countries in Europe to get a more detailed analysis.

As we can see in the graph in 2017 Poland had the highest cases of Cancer.

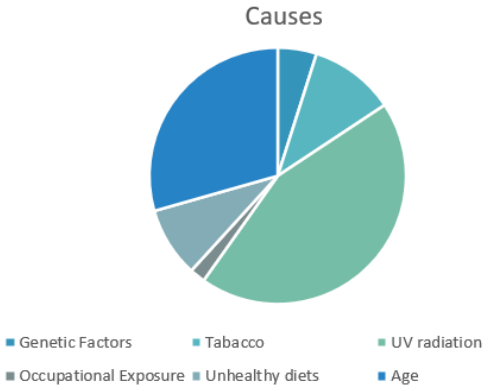
However, the maximum patients in the years that I have researched were Croatia in 1996 peaking at 989.835.739.776 cases in just one year being 197% higher than the average cases in those years.



Causes of cancer:

Some of the reasons for these evolutions over time and why it has increased so drastically over time is because of people's lifestyles nowadays. The amount of chemicals that have been added to our food, and different novice substances that we constantly put in our bodies such as alcohol and tobacco.

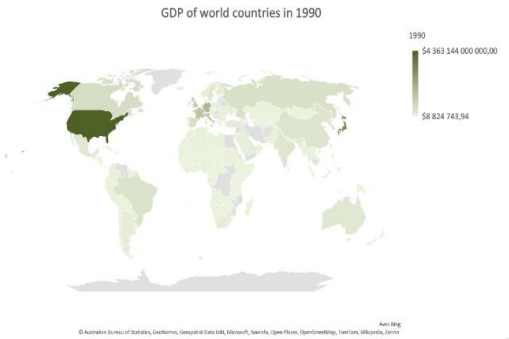
Another significant impact was the Chernobyl incident. This disaster was one of the most significant nuclear accidents in history. It had a significant impact because of the radiation that spread around the countries and increased the amount of cancer diagnosis over the years.

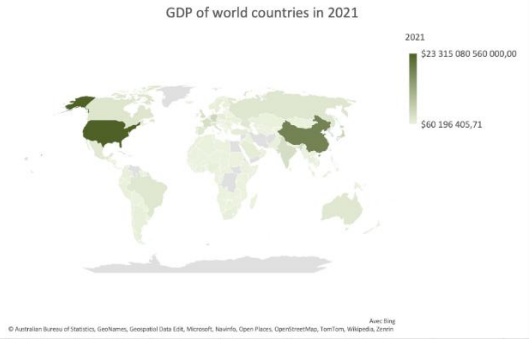


Cancer and link to economy

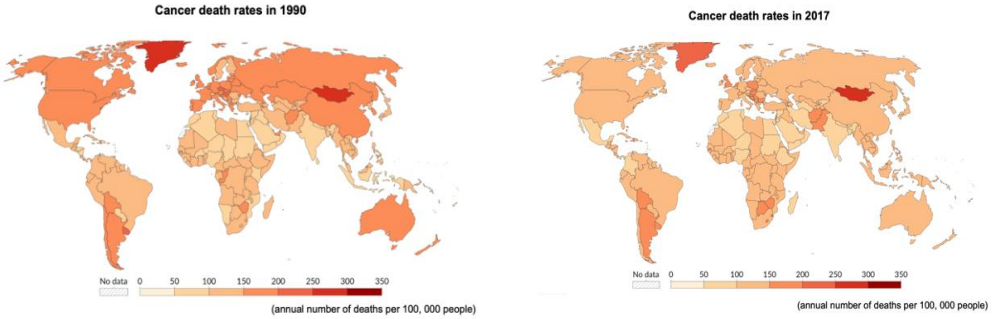
**Countries in grey do not provide information about their GDP.*

These two choropleth maps give us a picture of the GDP of world countries in the years 1990 and 2021 (most recent data). The GDP levels of countries give us an overview of which are the wealthiest countries in the world and consequently which are the ones with the most resources to combat cancer. The United States clearly dominates among the two years with its GDP being the highest of all during the 1990-2021 period. Throughout the years, new economies have emerged and strengthened, such as Japan, China, and some of the European countries. As for the lowest income countries, most of them are in Africa and South America.





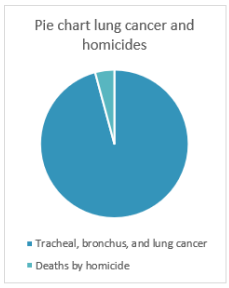
The two following maps present the cancer death rates in 1990 and 2017 (most recent data). These provide an oversight of the countries with the highest and lowest cancer death cases. Firstly, we can see that the 1990 map is generally more coloured than the 2017, meaning that worldwide cancer deaths were much higher back then. Although the general impression is that death rates have generally reduced, it is not the case for all world regions. In 1990, the most affected regions seemed to be Asia, North America, and Europe, and it is clear that these regions have successfully reduced their cancer death dates in the 2017 map. However, although the regions of South America, Africa and south Asia did not have particularly high cancer death rates in 1990, we can see that their rates have not really varied among the years and did not lower nor worsen for most countries in these regions.

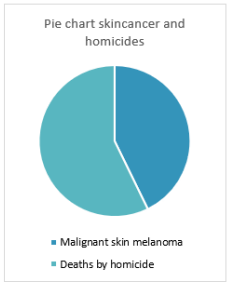


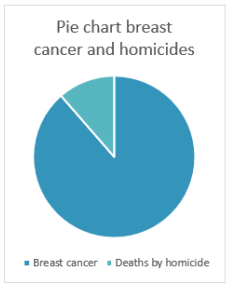
Overall, these maps allow us to compare the GDP and cancer deaths rate worldwide situation. I saw that cancer death rates have evolved and reduced throughout the years in the most developed world regions and that these have however stagnated in the rest of the world. In the same way, GDP levels have clearly increased in the regions of North America, Europe and Asia and other countries' GDP has generally stagnated between 1990 and 2021. I can therefore imagine that these two variables are correlated. The increase in GDP levels in the world's most developed regions led to a decrease in cancer death rates. Similarly, countries who maintained lower levels of GDP did not improve their cancer death rates.

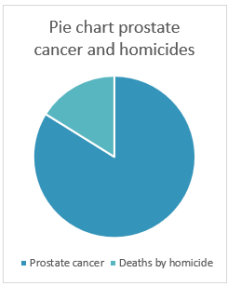
Cancer and other factors:

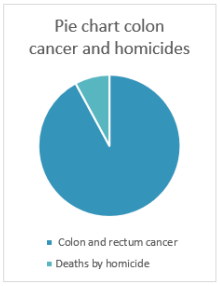
My last method of study was to establish quantitative continuous variables, which are variables produced by dividing one variable by another. The objective was to establish a percentage of the deaths by contrasting the number of deaths from cancer to the number of deaths from other causes. We divided the number of cancer deaths by the number of homicides, road fatalities, and alcohol and drug abuse deaths. I had to first calculate the arithmetic mean of the number of suicides per 100,000 people in the nations under study to divide one variable by the other. By doing this with our data, we got a variety of results the mean for breast cancer in 36.94, while for skin cancer in 3.33, the lung cancer mean is 104.82, the prostate cancer is 22.92 and lastly colon cancer's is 51.46. To present the results, we used the pie charts provided by excel.

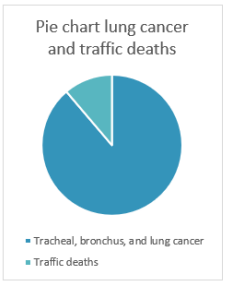


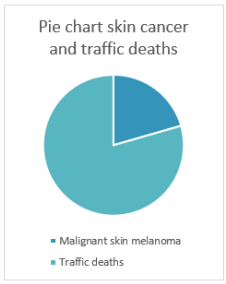


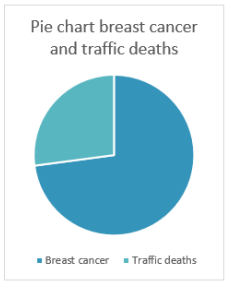


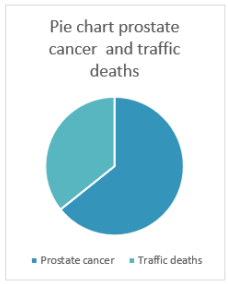


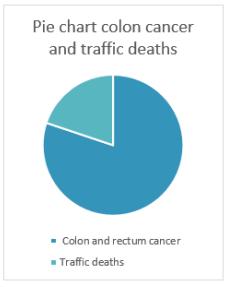


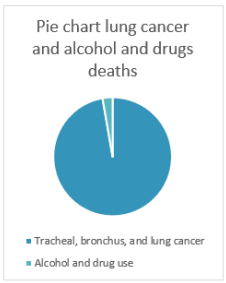


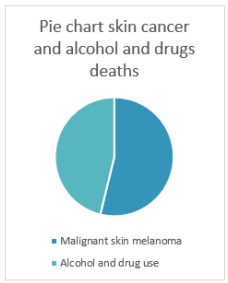


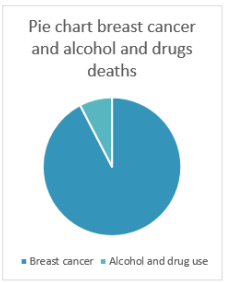


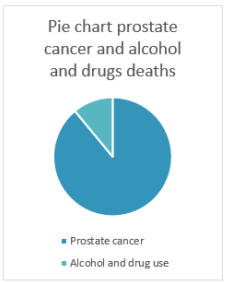


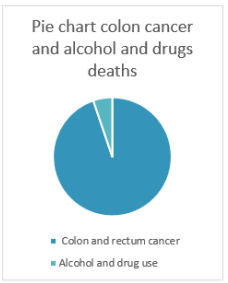












	Deaths by homicide	Traffic deaths	Alcohol and drug use
Tracheal, bronchus	22,86	7,91	35,60
Malignant skin mel	0,75	0,26	1,16
Breast cancer	7,77	2,69	12,11
Prostate cancer	5,20	1,80	8,09
Colon and rectum	11,66	4,04	18,16
Average	9,65	3,34	15,03

In Graphs 1 to 5, we can observe the proportion of deaths between the variables. In lung cancer and homicides, we can observe that there are 22,86 deaths by lung cancer for each death provoked by homicides, in malignant skin cancer the number drops to 0,75 deaths for each homicide, while in breast cancer there are 7,77 deaths for each homicide, for prostate cancer there are 5,2 for each death by murder, and lastly for there are 11,66 deaths by colon cancer for each death provoked by homicides. In Graphs 6 to 10, I evaluate the proportion of deaths between the diseases of study and the number of traffic deaths. In this case we can see that the numbers of death by type of cancer for each traffic death, decreases significantly. In the lung cancer case, that number passe to be 7,91 for each, in skin cancer it decreases to almost zero, being 0,26 deaths for each, in breast cancer it decreases to 2,69 deaths for each, prostate cancer are 1,8 deaths for each traffic death and for colon cancer the number is 4,04. Finally in graphs 11 to 15, it is analyse the proportion of deaths between the variables. In this case I analyse deaths by cancer ad deaths y alcohol and drugs. In lung cancer case the number is quite big, 35,6 deaths by cancer for each by addiction, in malignant skin cancer the number is 1,16 for each, in breast cancer is 12,11 per each, in prostate cancer occurs the biggest number being 8,09 deaths by the disease per each death by consumption, and lastly for colon cancer deaths are 18,16 deaths, for each by alcohol or drug abuse.

These findings demonstrate the seriousness of the cancer epidemic in our culture since cancer is the leading cause of mortality relative to other diseases and is more widely discussed on social media and in other forms of communication.

Conclusions and recommendations

In conclusion, during this research, I have found that the countries with most cancer cases are in Asia, Africa, and South America. It is essential that I am aware that many of my

daily activities nowadays are extremely unfavourable for my well-being, leading to health issues like cancer. Another thing I learned is that cancer is an excessively big issue around the world and not enough money is being used for research in this matter.

Secondly, I found out that cancer death rates are indeed correlated with the wealth of countries. The region with the biggest GDP increases over the years are also the ones which have reduced their cancer death numbers over the years. My hypothesis was therefore true: a wealthy country has much more resources available to combat and cure cancer, which results in lower cancer death rates.

Last, but not least, I found that the death rates from colon, skin, breast, prostate, and lung cancers worldwide surpass the death rates from drug consumption, traffic incidents, and homicides. This indicates that these cancers pose a more significant issue for my global society, causing numerous deaths each year. However, deaths from these cancers are not receiving as much attention as deaths from traffic incidents, for example.

APPENDIX

Glossary

THEORETICAL CONCEPTS

The definitions are taken from the Oxford dictionary;

- **GDP**: Gross Domestic Product. Total monetary value of all goods and services produced by a country. It is a key indicator used to measure the health of a country's economy.
- **Homicide**: the unlawful killing of one person by another.
- **Traffic deaths**: deaths that occur from road traffic accidents.
- **Drugs**: a medicine or other substance which has a physiological effect when ingested or otherwise introduced into the body
- **Alcohol**: a colourless volatile flammable liquid which is produced by the natural fermentation of sugars and is the intoxicating constituent of wine, beer, spirits, and other drinks, and is also used as an industrial solvent and as fuel
- **Wealth**: value of all the assets owned by a country.
- **Economy**: the process or system by which goods and services are produced, sold, and bought in a country or region.

EXCEL FUNCTIONS

- AVERAGE (number1, [number2], ...):
- STDEV.P (number1, [number2], ...):
- MAX (number1, [number2], ...):
- AVERAGE (A1; A...)

STATISTICAL CONCEPTS

- **Correlation coefficient:** Measure of the relative strength of the linear relationship between two variables.
- **Standard deviation**: measures the spread or dispersion of a set of values around the mean. It indicates how much individual data points deviate from the average value.

VISUALIZATION CONCEPTS

- **Pie chart (Excel):** A type of graph that represents data in a circular format. The slices of the pie illustrate the relative proportions of the data, serving as a visual representation of the information. It is better used to only two variables.
- **Map chart:** visual representation plotted on a map to show geographical patterns.
- Choropleth map: map which uses differences in shading/colouring to indicate the average values of a particular quantity in those areas.
- **Line graph:** a graph that shows how something has changed over time.
- **Bar graph:** graphical expression to show the data compared to distinct categories.

Workflow Chart

